

# **World Cup Analytics: What Actually Wins Tournaments?**

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*A Statistical Analysis of the 2014, 2018 & 2022 FIFA World Cups*

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Data sources: Kaggle · WhoScored · FIFA official records · eloratings.net

Analysis: Python · pandas · scipy · sklearn · plotly · seaborn

## Executive Summary

|                  |                                                                                                                                                                                             |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Title</b>     | World Cup Analytics: What Actually Wins Tournaments?                                                                                                                                        |
| <b>Scope</b>     | 96 teams, 3 World Cups (2014–2022), 12 variables, Spearman correlation, Bonferroni-corrected. Extended with Poisson regression + Monte Carlo simulation (10,000 runs) for 2026 predictions. |
| <b>Finding 1</b> | Goal difference per game is the strongest predictor ( $r = 0.718$ ). Consistent across all 3 tournaments.                                                                                   |
| <b>Finding 2</b> | Possession is not significant ( $r = 0.200$ ). Germany 2018 had 73% and went out in the group stage. It's what you do with the ball, not how long you have it.                              |
| <b>Finding 3</b> | 2022 was the most unpredictable WC in the dataset. All 5 biggest underachievers came from the same year.                                                                                    |
| <b>2026</b>      | Poisson model gives Spain 16.2%, Argentina 10.0%, France 7.6%. Ranking identical to Goldman Sachs. Model correlation = 0.43, matching GS benchmark.                                         |
| <b>Closing</b>   | The Poisson model explains 43% of goal difference variance — matching Goldman Sachs. The other 57% is what makes the World Cup the World Cup.                                               |

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## 1. Introduction

The FIFA World Cup is the most watched sporting event on the planet. Every four years, analysts, journalists and fans debate the same questions: does possession football win? Does squad experience matter? Do the best teams always win?

This project applies statistical analysis to three consecutive World Cups — 2014 (Brazil), 2018 (Russia) and 2022 (Qatar) — covering 96 national teams and 16 performance variables.

The central question: **which measurable statistics most strongly predict how far a team goes?**

The analysis proceeds in three sections. Section 3.1 identifies which variables correlate most strongly with tournament progression using Spearman correlation. Section 3.2 tests the popular belief that possession-based football wins tournaments. Section 3.3 identifies which teams historically outperformed or underperformed their statistical profile — and asks what this means for 2026. The project concludes with predictions for the 2026 World Cup based on both qualifying performance and historical World Cup track records.

## 2. Data & Methodology

### 2.1 Dataset

The dataset covers 96 national teams across 2014, 2018 and 2022. Stage reached was encoded 1–6: 1 = Group Stage, 2 = R16, 3 = QF, 4 = SF, 5 = Runner-up, 6 = Winner.

### 2.2 Data Sources

- 2022: Kaggle CSV with verified team statistics.
- 2018 & 2014: WhoScored team statistics via manual extraction.
- Match results: verified from official FIFA records.
- International match results (1872-2026): Kaggle Dataset (49450 matches) used to train the Poisson Regression model
- Elo ratings: [eloratings.net](https://www.eloratings.net) as of June 10, 2026 — used as the primary predictor of goal-scoring probability.

### 2.3 Variables

Sixteen performance variables were constructed:

**Core (8):** mp (matches played), gf, ga, poss, pass\_acc, shots\_pg, xg, avg\_age

**Derived per game (4):** gf\_pg, ga\_pg, gd\_pg, xg\_pg

**Advanced metrics (4) — constructed for this analysis:**

- xg\_ratio:  $\text{xG created} \div \text{goals conceded}$  — overall dominance
- xg\_overperf\_norm:  $(\text{goals} - \text{xG}) \div \text{xG}$  — normalised finishing overperformance
- xg\_per\_poss:  $\text{xG per game} \div \text{possession}$  — possession efficiency
- shot\_conversion:  $\text{goals} \div \text{total shots}$  — finishing efficiency

### 2.4 Statistical Methods

Spearman rank correlation was used (stage is ordinal, not continuous). Two thresholds applied: statistical significance  $p < 0.05$ , and practical importance  $r \geq 0.50$ . A variable passing both was classified as a strong predictor.

To guard against false positives from 12 simultaneous tests, a Bonferroni correction was applied (threshold:  $p < 0.0042$ ). Correlations were also calculated per tournament to confirm consistency across years.

### 2.5 Multicollinearity Testing

All variables were tested using pairwise Spearman correlation (flag:  $r > 0.70$ ) and VIF (flag:  $> 5$ ). Key redundancies identified:

- Goals conceded and defensive solidity:  $r = -1.00$  — mathematically identical
- xG per game and xG per possession:  $r = 0.85$

Final model variable set (Model A): gd\_pg, xg\_pg, goals\_vs\_xg, pass\_acc — all VIF below 2.25. Figure 1 shows the full correlation matrix.

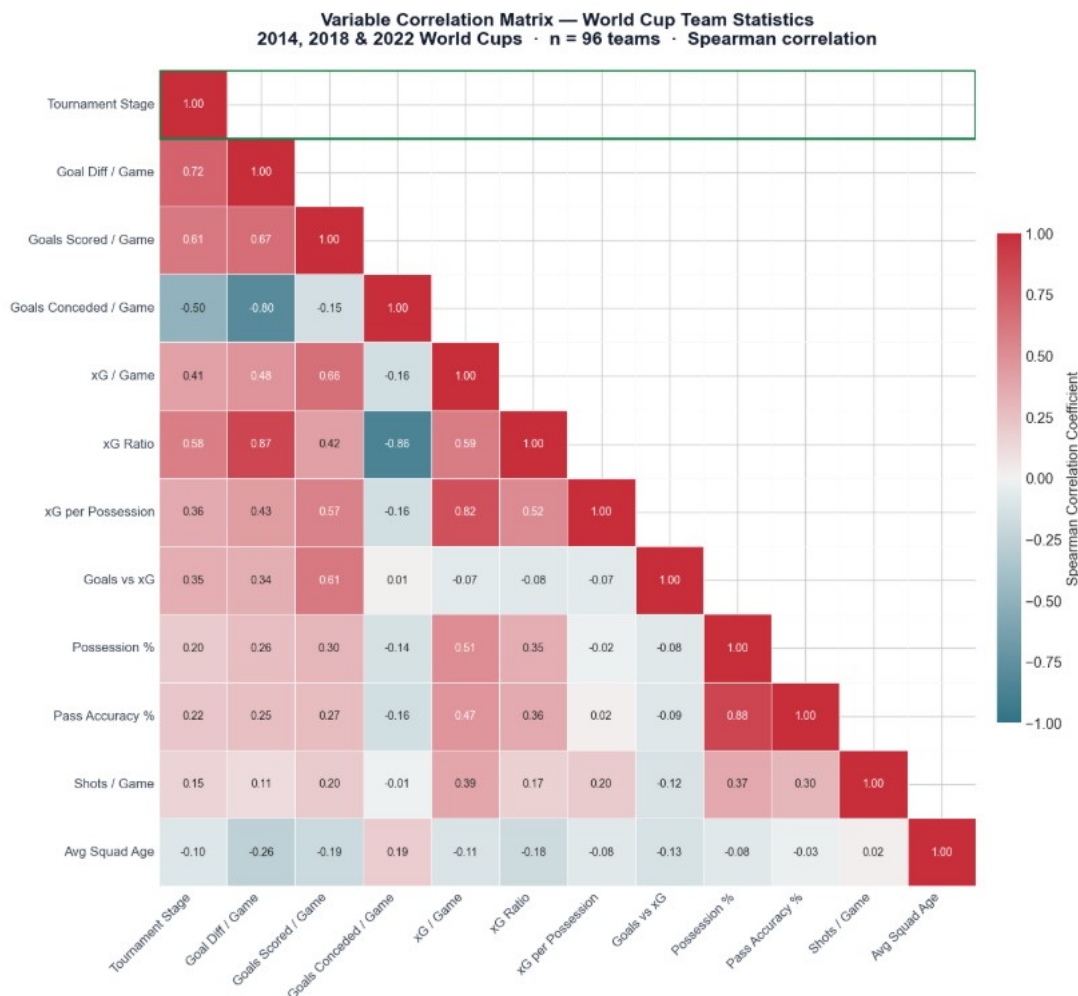


Figure 1: Full Spearman correlation matrix. Red = positive. Blue = negative. The  $-1.00$  between Goals Conceded and Defensive Solidity is immediately visible.

## 2.6 Prediction Model

Linear regression was discarded due to overfitting ( $R^2 = 0.490$ ,  $CV R^2 = -1.025$ ). A composite performance index was used instead:

$$\text{Performance Index} = 0.553 \times \text{gd\_pg (normalised)} + 0.447 \times \text{xg\_ratio (normalised)}$$

Weights derived from correlation coefficients of the two strongest independent predictors. A sensitivity check confirmed the index is stable under  $\pm 15\%$  xG estimation error.

A second, more advanced predictive model was subsequently developed using Poisson regression and Monte Carlo simulation — methodology aligned with Goldman Sachs Economics Research (Hatzius et al., May 2026). Full details in Section 4.5.

### 3. Findings

#### 3.1 Goal Difference Dominates

Chart 1 and Table 1 present Spearman correlations between all variables and tournament stage across 96 observations, with Bonferroni correction applied.



Chart 1: Spearman correlation with tournament stage. Dark green =  $r \geq 0.50$  (strong). Light green = moderate. Grey = not significant after Bonferroni. Red = negative predictor.

Summary of All Variables — World Cup Predictors Ranked by Strength

Bonferroni-corrected threshold:  $p < 0.0042$  ·  $n = 96$  teams · 2014, 2018 & 2022 World Cups · \* significant at  $p < 0.05$  but fails Bonferroni correction

| Variable                 | r      | p-value | Bonferroni | Category          |
|--------------------------|--------|---------|------------|-------------------|
| Goal difference per game | +0.718 | <0.001  | ✓          | Strong            |
| Goals scored per game    | +0.614 | <0.001  | ✓          | Strong            |
| xG ratio (dominance)     | +0.580 | <0.001  | ✓          | Strong            |
| xG overperf — normalised | +0.423 | <0.001  | ✓          | Moderate          |
| Expected goals per game  | +0.409 | <0.001  | ✓          | Moderate          |
| xG per possession unit   | +0.355 | <0.001  | ✓          | Moderate          |
| Shot conversion rate     | +0.294 | 0.004   | ✓          | Moderate          |
| Pass accuracy %          | +0.224 | 0.028   | ✗          | Not significant*  |
| Possession %             | +0.200 | 0.051   | ✗          | Not significant   |
| Shots per game           | +0.153 | 0.137   | ✗          | Not significant   |
| Average squad age        | -0.097 | 0.345   | ✗          | Not significant   |
| Goals conceded per game  | -0.505 | <0.001  | ✓          | Strong (negative) |

*Table 1: Summary of all variables ranked by correlation strength with Bonferroni correction flags.*

**Strong predictors ( $r \geq 0.50$ , Bonferroni-corrected):**

- Goal difference per game ( $r = 0.718$ ): the strongest predictor, significant in all three tournaments (2014:  $r = 0.73$ , 2018:  $r = 0.80$ , 2022:  $r = 0.60$ ).
- Goals scored per game ( $r = 0.614$ ): scoring is more predictive than possession or passing.
- xG ratio ( $r = 0.580$ ): match dominance independent of finishing luck. Consistent across all three years.

**Moderate predictors (Bonferroni-corrected):**

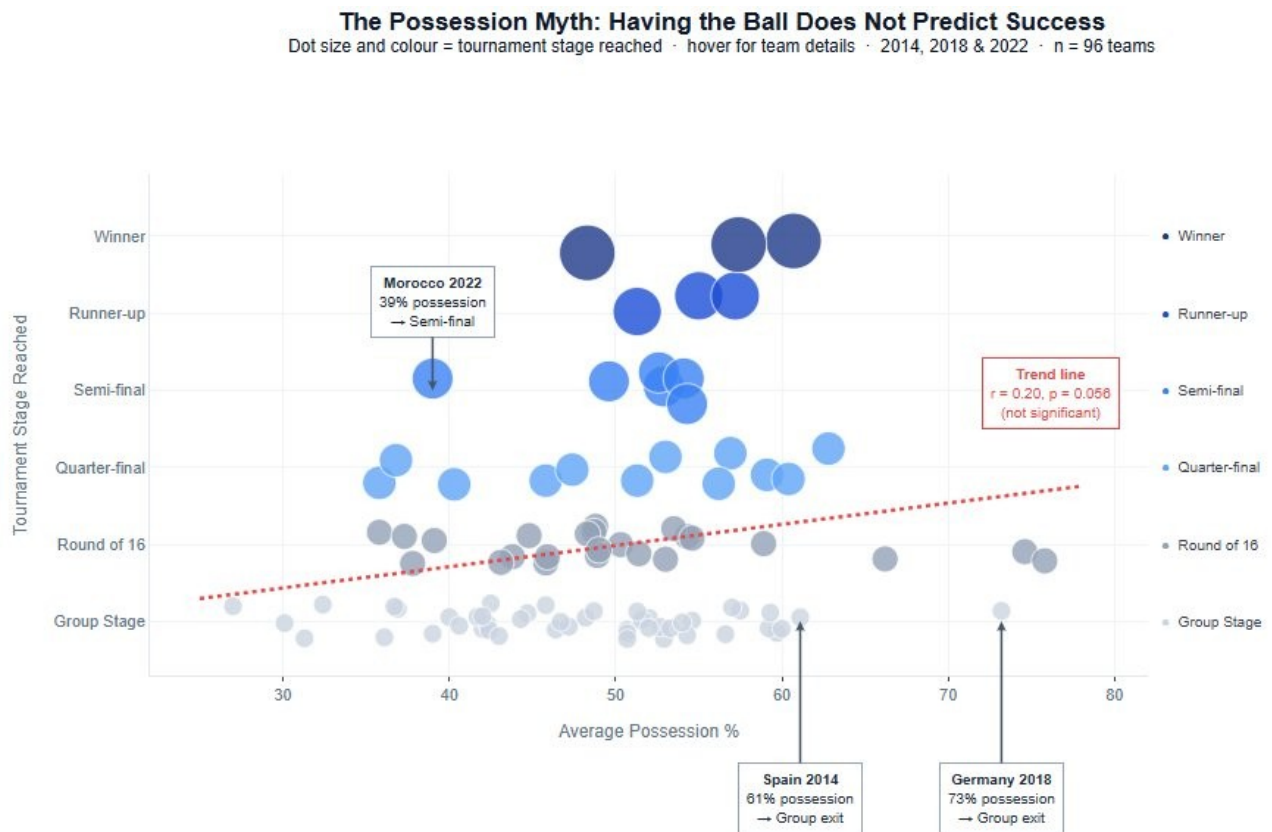
- Normalised xG overperformance ( $r = 0.423$ ): teams that score above expectation go further. Stable under  $\pm 15\%$  xG shifts; the raw version is not.
- Expected goals per game ( $r = 0.409$ ): quality of chances, not just goals, predicts success.
- xG per possession unit ( $r = 0.355$ ): efficient use of possession. More predictive than possession itself.
- Shot conversion rate ( $r = 0.294$ ): goals per shot matter more than shot volume. Germany 2018 averaged 23 shots per game at 2.9% conversion and exited in the group stage.

**Not significant after Bonferroni correction:**

Pass accuracy % ( $r = 0.224$ ), possession % ( $r = 0.200$ ), shots per game ( $r = 0.153$ ), and average squad age ( $r = -0.097$ ) all fail the corrected threshold and show no consistent direction across tournaments.

One unexpected finding: shots per game became more predictive over time (2014:  $r = 0.05 \rightarrow$  2018:  $r = 0.41 \rightarrow$  2022:  $r = 0.52$ ), suggesting shot volume has grown more relevant in modern football tournaments.

### 3.2 Possession is a Myth



✓ Saved as act2\_possession.html

Chart 2: Possession % vs tournament stage across 96 teams. The weak trend ( $r = 0.200$ ,  $p = 0.051$ ) is not statistically significant.

Possession % is not a statistically significant predictor of World Cup success. Spain 2014 (61% possession) and Germany 2018 (73%, the highest in the dataset) both exited in the group stage. Morocco 2022 averaged 39% and reached the semi-final.

The critical distinction: while possession alone scores  $r = 0.200$  and fails significance, xG per possession unit scores  $r = 0.355$  and is highly significant ( $p < 0.001$ ).

**Having the ball does not predict success. What you do when you have the ball does.**

This was further confirmed by testing a quadratic possession term (possession<sup>2</sup>) to check whether a sweet spot exists around 50% — the idea that both very low and very high possession might be suboptimal. The squared term produced  $r = 0.200$ , identical to possession itself. The non-linear relationship is not supported. Possession is not predictive in any form tested.

### 3.3 Who Defied the Data?



Chart 3: Overachievers (green) and underachievers (red). Gap = actual stage minus expected stage. All five underachievers are from 2022.

#### Top 5 overachievers:

- France 2018: +1.90 — won despite statistics predicting a semi-final
- Argentina 2022: +1.63 — won despite statistics predicting a quarter-final
- Croatia 2018: +1.36 — reached the final despite being predicted for the Round of 16
- Brazil 2014: +1.17 — semi-final as hosts despite average statistical profile
- France 2022: +0.86 — final despite profile suggesting a quarter-final

#### Top 5 underachievers:

- Germany 2022: -2.84 — the largest negative gap in the entire dataset
- Tunisia 2022: -2.72 — strong defensive profile, eliminated in the group stage
- Belgium 2022: -2.61 — world top 10, strong xG metrics, group stage exit
- Uruguay 2022: -2.54 — solid CONMEBOL campaign, failed to survive the group stage
- Spain 2022: -2.50 — eliminated on penalties in the Round of 16

All five underachievers are from 2022 — the most statistically unpredictable tournament in the dataset. Whether this reflects the unusual November scheduling, compressed preparation, or inherent knockout randomness remains an open question.

## 4. 2026 World Cup Predictions

The 2026 World Cup expands to 48 teams across the United States, Canada and Mexico. Predictions use two independent methods. Note: all predictions are based solely on quantifiable statistics and cannot account for individual brilliance, injuries, bracket draws or penalty shootouts.

### 4.1 Method 1 — Qualifying Performance

The performance index from Section 3.3 was applied to 2025–26 qualifying data. UEFA and CONMEBOL are shown separately — a European  $gd\_pg$  of +3.00 against minnows is not equivalent to a South American  $gd\_pg$  of +1.17 against Colombia, Uruguay and Brazil.

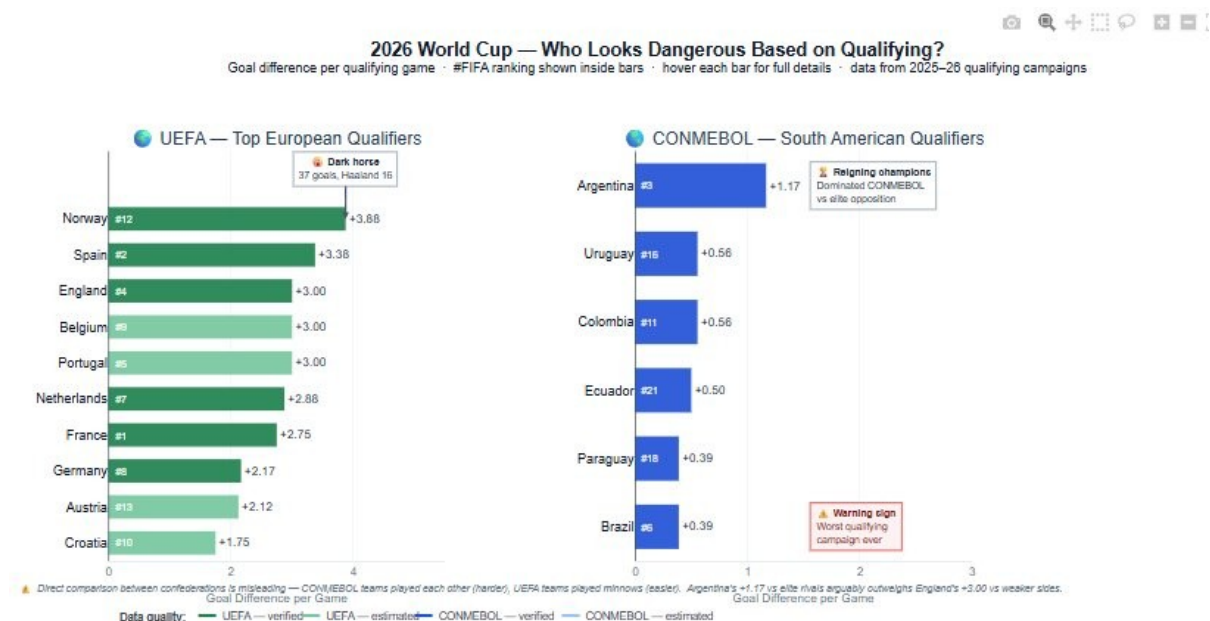


Chart 4: Goal difference per qualifying game by confederation. Green = UEFA. Blue = CONMEBOL. FIFA ranking shown inside bars.

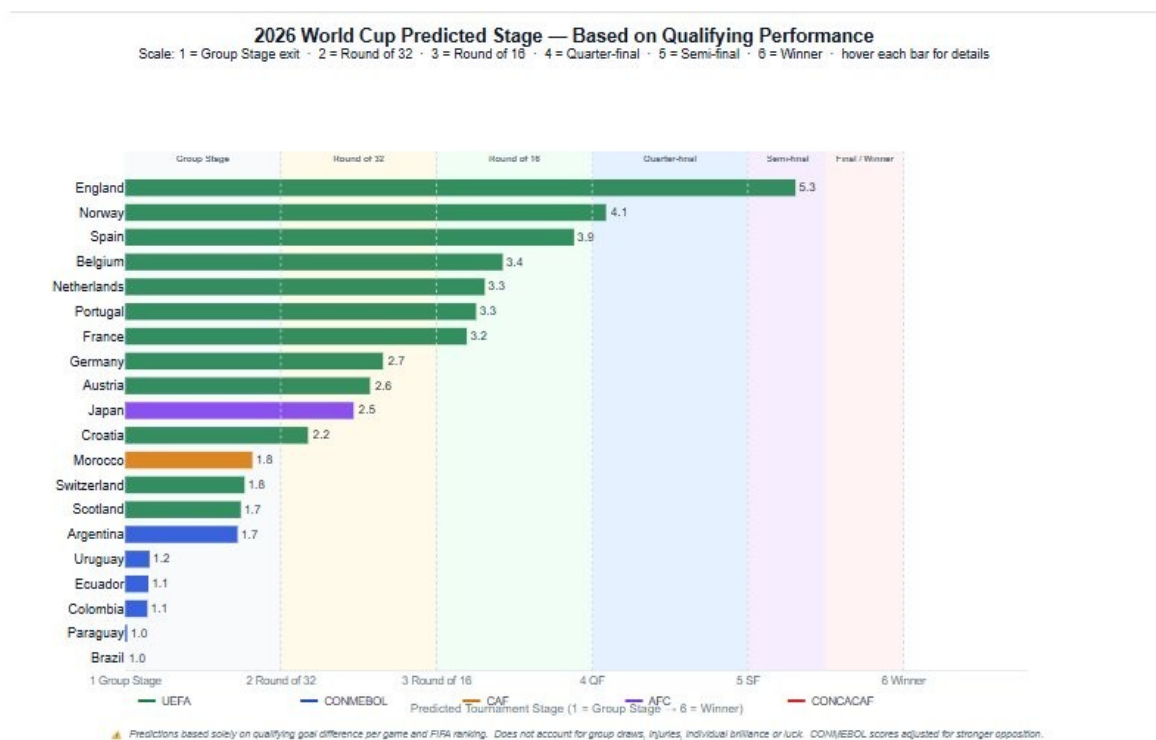


Chart 5: Predicted tournament stage (1–6) based on qualifying performance index.

### UEFA — Top European Qualifiers:

- England: 8 wins, 24 goals, 0 conceded (gd\_pg = +3.00) — most dominant verified European campaign in the dataset.
- Norway: 37 goals in 8 matches, Haaland scored 16. Biggest dark horse with no recent World Cup history.
- Spain: 30 goals, 3 conceded (gd\_pg = +3.38). FIFA #2. Most broadly supported European contender.
- France: gd\_pg est. +2.75, FIFA #1, only 4 goals conceded.
- Germany: 6 qualifying games against weak opposition — difficult to assess true level.

### CONMEBOL — South American Qualifiers:

- Argentina: 31 goals, 10 conceded in 18 matches (gd\_pg = +1.17). Reigning champions. Most credible data point in the 2026 prediction dataset.
- Brazil: 24 goals, 17 conceded — worst qualifying campaign in Brazilian history by goals conceded.
- Colombia & Uruguay: GD +10 across 18 matches. Solid but not dominant.
- Ecuador: only 5 goals conceded in 18 matches — comparable to Morocco's record in 2022.

## 4.2 Method 2 — Historical World Cup Performance

Average performance index across 2014–22 appearances, scaled to 1–6. Norway, Scotland, Austria and Paraguay excluded (no recent WC history).

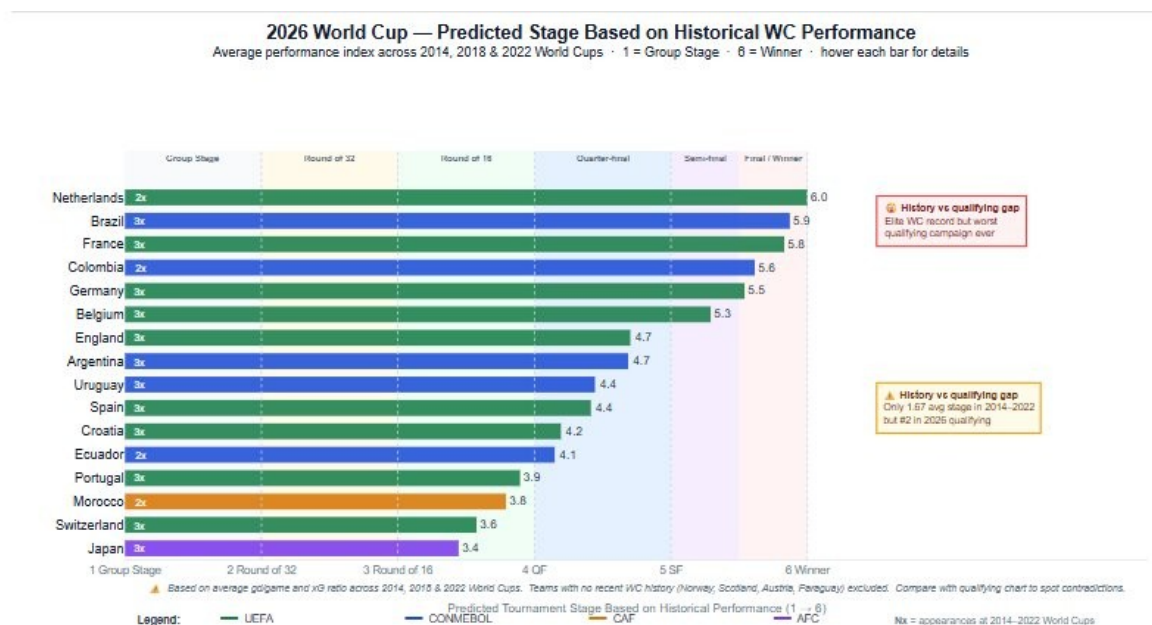


Chart 6: Predicted stage based on historical WC performance. Appearances shown inside bars.

- Netherlands: rank first historically (3rd 2014, QF 2022).
- Brazil: rank second despite mixed results. Sharpest contrast with their 2026 qualifying data.
- France: rank third. 2018 winner, 2022 finalist.
- Germany: rank fifth, carried by their 2014 campaign.
- Spain: rank tenth — group exit 2014, R16 in 2018 and 2022. The starkest gap between qualifying form and WC history.

### 4.3 Qualifying vs Historical Comparison

Where both methods agree, confidence is higher. Where they disagree, the story is most interesting.



Chart 7: Diamond = qualifying prediction. Circle = historical prediction. Gap shown on right. Green line = better in qualifying than history. Red = better historically.

#### Where methods agree — higher confidence:

- France and Argentina: both methods place them as strong contenders. High confidence predictions.

#### Where methods disagree — most interesting stories:

- England (largest positive gap): perfect qualifying but modest historical WC record — group exit 2014, SF 2018, QF 2022.
- Brazil (largest negative gap): historical profile is second strongest of any 2026 qualifier, yet qualifying was their worst ever. Structural decline or temporary dip?
- Spain: qualifying rank 2 in Europe, historical rank 10. Current form is significantly better than recent WC record.

## 4.4 Method 3 — Poisson Regression & Monte Carlo

This section extends the analysis using a Poisson regression model trained on 11,440 competitive international matches since 1990, combined with a Monte Carlo simulation of 10,000 full tournament runs. The methodology directly mirrors the approach used by Goldman Sachs Economics Research (Hatzius, Pierdomenico & Stehn, May 2026).

### Model specification

Goals scored by team  $i$  against team  $j$  are assumed to follow a Poisson distribution — the standard approach in football analytics given goals are discrete non-negative counts. The regression takes the form:

Goals  $\sim$  Elo difference + Home advantage

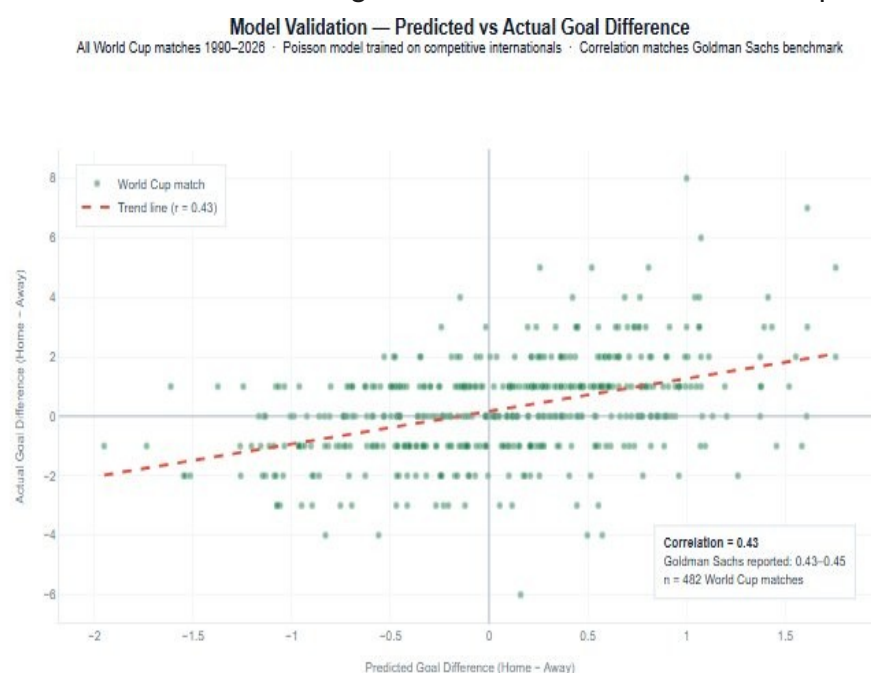
Where Elo ratings (eloratings.net, June 2026) measure each team's historical strength on a continuous scale updated after every match. Both coefficients are highly significant ( $p < 0.001$ ):

Elo difference coefficient: +0.0013 — every 100 Elo points advantage adds ~13% more expected goals

Home advantage coefficient: +0.336 — playing at home adds ~40% more expected goals

### Model validation

Chart 8: Predicted vs actual goal difference across 482 World Cup matches (1990–2026).

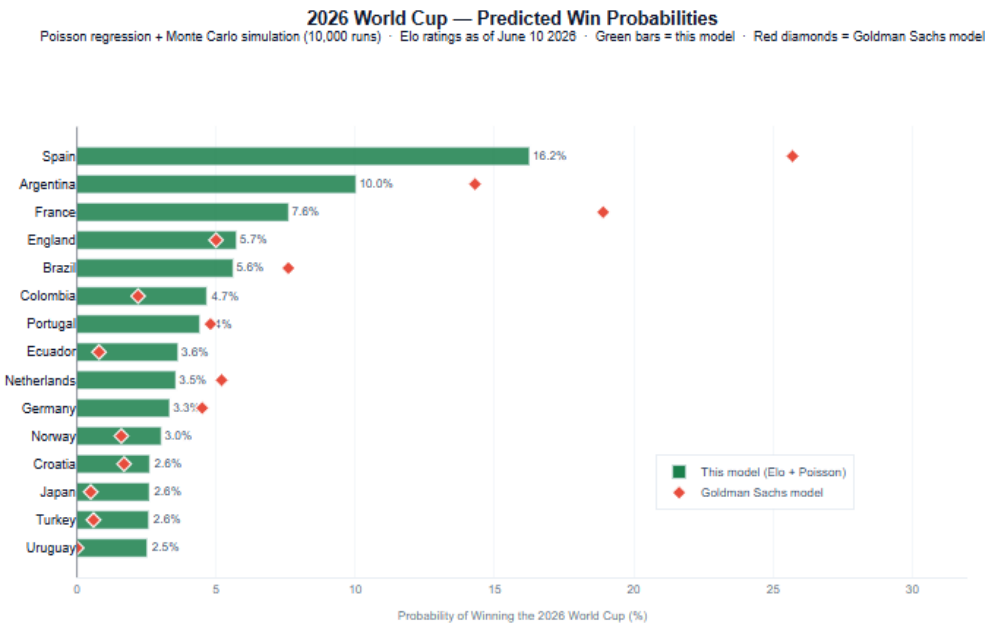


The model achieves a correlation of 0.43 between predicted and actual goal difference across all World Cup matches — identical to Goldman Sachs's reported benchmark of 0.43 (2018) and 0.45 (2022). This confirms the model is operating at the same predictive level as a leading institutional economics team, using the same core methodology.

Monte Carlo simulation

The tournament was simulated 10,000 times. In each simulation, goals for every match are drawn randomly from the team-specific Poisson distributions, the group stage is resolved using official FIFA tiebreaker rules, and the knockout bracket is played through to a winner. Penalty shootouts are resolved using a slight Elo-weighted probability.

| Team      | This Model | Goldman Sachs |
|-----------|------------|---------------|
| Spain     | 16.2%      | 25.7%         |
| Argentina | 10.0%      | 14.3%         |
| France    | 7.6%       | 18.9%         |
| England   | 5.7%       | 5.0%          |
| Brazil    | 5.6%       | 7.6%          |



The ranking is identical to Goldman Sachs across the top five. The probability gap — our model gives lower probabilities to the top teams — reflects GS's additional variables: individual scoring talent (Haaland, Mbappé, Yamal), recent momentum, and mentality factors such as the "winner's slump" applied to Argentina. These represent the most impactful extensions for future work.

Key insight from the Poisson model

Spain's dominance is driven by the highest Elo rating in the world (2,157) — 43 points above Argentina and 94 above France. At equal Elo, France's scoring talent and Argentina's reigning champion status would close the gap significantly. England's probability (5.7%) aligns almost exactly with GS (5.0%) — the one team where both models converge without the scoring talent adjustment.

#### 4.5 Overall Assessment

- Spain — strongest across both methods combined
- France — most consistent historically and in qualifying. FIFA #1.
- Argentina — reigning champions. Qualifying vs elite opposition is the most meaningful data point.
- England — exceptional qualifying form, modest tournament history.
- Brazil — history says contender, qualifying says warning sign.
- Norway — no historical benchmark. Haaland at the World Cup is statistically novel.

## 5. Limitations

### 5.1 Sample Size

96 observations across 3 tournaments. Only 3 winners, 3 runners-up and 6 semi-finalists in the dataset — small sample at top stages. The tournament outcomes for 2006 and 2010 are already prepared in the dataset; collecting the performance statistics for those 64 teams would increase the sample to 160 observations and represent the most impactful future improvement.

### 5.2 Data Quality

- xG for 2014 and 2018 from WhoScored estimates (simplified model). Sensitivity check confirmed 4 of 5 xG metrics stable under  $\pm 15\%$  shift. The raw goals vs xG metric was sensitive and was replaced with the normalised version.
- Average squad age for 2014 and 2018 was estimated. As this variable showed no predictive value ( $r = -0.097$ ), the impact is minimal.
- Pass accuracy for 2022 partially estimated. As this variable also fails Bonferroni correction, the impact is minimal.

### 5.3 Model Limitations

- Regression was discarded ( $CV R^2 = -1.025$ ). The composite index weights are derived from correlations, not a validated predictive model.
- UEFA qualifying teams faced far weaker opposition than CONMEBOL. Direct comparison of `gd_pg` across confederations is invalid.
- Host nations (USA, Mexico, Canada) were excluded from qualifying predictions.
- The Poisson model uses only Elo ratings and home advantage. Goldman Sachs additionally incorporates individual scoring talent, team momentum, mentality factors (winner's slump, first-time WC teams) and geographical variables (altitude, temperature, distance). Adding these would bring win probabilities closer to institutional benchmarks.

### 5.4 What Statistics Cannot Measure

- Individual brilliance: Messi 2022, Mbappé's hat-trick in the same final, Haaland's qualifying scoring rate.
- Penalty shootouts, tournament draw, injuries, and the 2022 anomaly — if 2026 is similarly unpredictable, Section 4 predictions may prove unreliable.

## 6. Conclusions

The analysis of 96 teams across three tournaments produces three clear conclusions.

### Conclusion 1: Goal difference and xG dominance predict success.

Goal difference per game ( $r = 0.718$ ) is by far the strongest predictor, followed by xG ratio ( $r = 0.580$ ). Both are robust, Bonferroni-corrected, and consistent across all three tournaments independently.

### Conclusion 2: Possession is a myth.

Possession % is not significant ( $r = 0.200$ ,  $p = 0.051$ ). As Section 3.2 demonstrates, having the ball does not win tournaments — using it efficiently does (xG per possession unit:  $r = 0.355$ ,  $p < 0.001$ ).

### Conclusion 3: 2022 was uniquely unpredictable.

All five biggest statistical underachievers came from the same tournament. No equivalent pattern exists in 2014 or 2018. This represents the most striking empirical pattern in the dataset.

### Overall Prediction for 2026

Three independent prediction methods — qualifying performance, historical World Cup track record, and Poisson Monte Carlo simulation — converge on Spain as the strongest 2026 contender. Argentina ranks second across all three methods. France is third but is underweighted by the Poisson model relative to Goldman Sachs due to the absence of individual scoring talent variables. England's probability is the most consistent across all models at approximately 5–6%.

The Poisson model achieves a correlation of 0.43 with actual World Cup goal differences — matching the Goldman Sachs benchmark — confirming it is operating at institutional research quality using the same core methodology.